Simulation Of The Flow Over A Backward-Facing Step

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Abstract

The LES code in OpenFOAM is used to simulate the flow over a backward-facing step using three different mesh sizes. The maximum pressure values of the three different cases are compared and a mesh convergence study is carried out.

List of Symbols

α	Phase fraction of the given phase
ϵ	Turbulent kinetic energy dissipation rate $[m^2/s^3]$
ν	Kinematic viscosity $[m^2/s]$
ν_t	Turbulent viscosity $[m^2/s]$
ρ	Density of the fluid $[kg/m^3]$
C_1, C_2, C_3	Model coefficients
C_μ	Model coefficient for the turbulent viscosity
$C_{3,RDT}$	Turbulent kinetic energy production rate due to the anisotropic part of the Reynolds-stress tensor $[m^2/s^3]$
D_ϵ	Effective diffusivity for ϵ
D_k	Effective diffusivity for k
G	Turbulent kinetic energy production rate due to the anisotropic part of the Reynolds-stress tensor $[m^2/s^3]$
I	Turbulence intensity [%]
k	Turbulent kinetic energy $[m^2/s^2]$
l	A reference length scale $[m]$
N_x	Number of cells in the x direction
N_y	Number of cells in the y direction
N_z	Number of cells in the z direction
P	Turbulent kinetic energy production rate $[m^2/s^3]$
p	Kinematic pressure $[m^2/s^2]$
R	Viscous stress term
$S_{fvOptions}$	Source terms introduced by fvOptions dictionary for $\epsilon and k$
S_ϵ	Internal source term for ϵ
S_k	Internal source term for k
U	Velocity vector $[m/s]$
u_{ref}	A reference flow speed $[m/s]$

1. Introduction

Numerical methods are sometimes misleading since they might generate solutions that do not converge to the analytical solution; several techniques are used to check the convergence of solutions. One of them is the mesh convergence method. This method is universal and can be used even in cases where neither the analytical solution to the problem at hand is known nor experimental data are available to be compared against the solution obtained.

This study aims to apply the mesh convergence method to check the convergence of the simulation of air inside a duct with a backward-facing step. This simulation replicates an experiment executed by Robert W.Pitz and John W.Daily in which they assessed the effect of combustion on the mean flow field properties such as mixing layer growth, entrainment rate, and reattachment length. This report focuses on studying the turbulent mixing layer formed inside the duct.

2. Theory

In this study the open source software OpenFOAM was used. To simulate the Pitz and Daily experiment a number of assumptions are made, the flow is taken to be turbulent, viscous, two-dimensional, incompressible, and transient. Hence, computations are performed using the transient solver for

incompressible flows available in OpenFOAM (PISOFoam).It uses the PISO algorithm for pressure–velocity decoupling.

2.1. Governing Equations

The aforementioned assumptions are translated into the following governing equations:
1. Incompressibility, given by the continuity equation:

$$\nabla.U = 0 \tag{1}$$

2.The transient momentum equation, given by:

$$\frac{\partial u}{\partial t} + \nabla.(u \otimes u) = -\nabla p \tag{2}$$

Initial and boundary conditions (given in table(1)) should be added to equations (1) and (2).

Initial Conditions	Boundary Conditions
$U = 0$ m/s	$U_{inlet} = (10, 0, 0)$ m/s
$p = 0$ Pa	$p_{outlet} = 0$ Pa

Table 1: Initial and boundary conditions for equations (1) and (2).

2.2. Turbulence Modelling

The PISOFoam solver-which uses Large Eddy Simulation (LES)- is used with the famous $k - \epsilon$ turbulence model, the model uses two equations, one for the turbulent kinetic energy (k), and one for the rate of dissipation of that turbulent kinetic energy ϵ as follows:

$$\frac{D}{Dt}(\rho k) = \nabla.(\rho D_k \nabla k) + P - \rho\epsilon \tag{3}$$

$$\frac{D}{Dt}(\rho\epsilon) = \nabla.(\rho D_\epsilon \nabla\epsilon) + \\ + \frac{C_1\epsilon}{k}(P + C_3\frac{2}{3}k\nabla.u) - C_2\rho\frac{\epsilon^2}{k} \tag{4}$$

Those two equations (3) and (4) are implemented in openFOAM as follows:

The turbulent kinetic energy equation (k):

$$\frac{\partial}{\partial t}(\alpha\rho k) + \nabla.(\alpha\rho u k) - \nabla^2(\alpha\rho D_k k) = \alpha\rho G - \\ -(\frac{2}{3}\alpha\rho\nabla.uk) - (\alpha\rho\frac{\epsilon}{k}k) + S_k + S_{fvOptions} \tag{5}$$

The turbulent kinetic energy dissipation rate (ϵ):

$$\frac{\partial}{\partial t}(\alpha\rho\epsilon) + \nabla.(\alpha\rho u\epsilon) - \nabla^2(\alpha\rho D_\epsilon\epsilon) = C_1\alpha\rho G\frac{\epsilon}{k} - \\ -((\frac{2}{3}C_1 - C_{3,RDT})\alpha\rho\nabla.u\epsilon) + S_\epsilon + S_{fvOptions} \tag{6}$$

The coefficients appear in the aforementioned equations are summarized in table(2).

Coefficient	Value
C_μ	0.09
C_1	1.44
C_2	1.92
α_k	1
α_ϵ	0.76

Table 2: $k - \epsilon$ turbulence model coefficients.

An important concept used in the $k - \epsilon$ model is the turbulent viscosity, given by:

$$\nu_t = C_\mu\frac{k^2}{\epsilon} \tag{7}$$

To solve equations (5) and (6), they initial values for k and ϵ should be given.

1. For k, the initial value is calculated using the following equation:

$$k = \frac{3}{2}(I|u_{ref}|)^2 \tag{8}$$

If we assume that the inlet turbulence is isotropic and the fluctuations to be 5% ,we get:

$$U_x{}' = U_y{}' = U_z{}' = \frac{5}{100}10 = 0.5 \ \ m/s$$

substituting back in equation (8), we find:

$$k = \frac{3}{2}(0.5)^2 = 0.375 \ \ m^2/s^2$$

2. For ϵ, the initial value is calculated using:

$$\epsilon = \frac{{C_\mu}^{0.75}k^{1.5}}{l} \tag{9}$$

using C_μ value from table (2), and if the turbulent length scale (l) is estimated to be 10% of the width of the inlet then:

$$\epsilon = \frac{0.09^{0.75} \times 0.375^{1.5}}{0.1 \times 25.4 \times 10^{-3}} = 14.855 \ \ m/s \tag{10}$$

2.3. Mesh Convergence

To check whether the numerical results obtained will converge to the analyical solution or not, a number of techniques are used, one of them is the mesh convergence method. The first step in this method is to calculate the Grid Convergence Index (GIC) for any two grids is calculated using:

$$GCI_{2,1} = \frac{\frac{F_2 - F_1}{F_1}}{r^O - 1} \tag{11}$$

The variable O that appears in this equation is known as the order of convergence and is calculated by:

$$O = \frac{\ln \frac{F_3 - F_2}{F_2 - F_1}}{\ln(r)} \tag{12}$$

the convergence of the numerical scheme can be checked using the following formula:

$$GCI_{3,2} \approx r^O GCI_{2,1} \tag{13}$$

or in other words, $GCI_{3,2}$ divided by $GCI_{3,2}$ should approximately equal r^O

3. Results and Discussion

Three different mesh sizes are used in this study given in table(3) and shown in figure (2). From the mesh convergence study results, we see that $\frac{GCI_{3,2}}{GCI_{3,2}} = 1.276$ and $r^O = 1.175$, which means that the difference between the two values is 8.5% which is acceptable.

Grid	N_x	N_y	N_z
1	284	82	1
2	426	138	1
3	639	207	1

Table 3: Size of different grids used in the study.

3.1. Mean Values

From the velocity values (figure (3)), one can see that the three different grid sizes generated almost identical results. At the inlet, velocity values are high. Due to the sudden change in the duct shape, flow separation occurs in the step area, see figure (5). Hence circulation forms. The main flow feeds this circulation. Thus the velocity of the main flow decreases considerably. Before reaching the outlet, the flow reattaches, then gradually recovers part of the lost velocity.

Pressure values are shown in figure (4). According to Bernoulli's equation, the pressure values are high where the velocity is low and low where the velocity is high.

3.2. The Underlying Mechanism

It is known that average values show the dominating phenomena in the flow, yet they fail to give the details of the creation and distortion of those phenomena. For example, streamlines of the mean velocity (figure(5)) shows only one zone in which circulation is developed, while when one looks at figure (6), they see that there are two zones.

The underlining mechanism is as follows. When the flow reaches the step area, it separates, and part of it flows downwards to fill the void creating a small bubble. This bubble keeps growing until it reaches a certain size, then the bubble moves downstream, giving space for a new bubble to be created. See figures (6(a)) to (6(c)). The movement of the aforementioned bubble creates another void upwards. Later on, this void is filled by a new bubble. This bubble also keeps traveling in the downstream direction. This bubble is created at a lower rate when compared to the second bubble.

4. References

1. https://www.openfoam.com/documentation/ tutorial-guide/3-compressible-flow/3.1-steady -turbulent-flow-over-a-backward-facing-step.

2. J.Russell, Pitz-Daily Turbulence Case, https://www.bu.edu.

3.https://www.openfoam.com/documentation /guides/latest/doc/guide-turbulence-ras-k-epsilon.html.

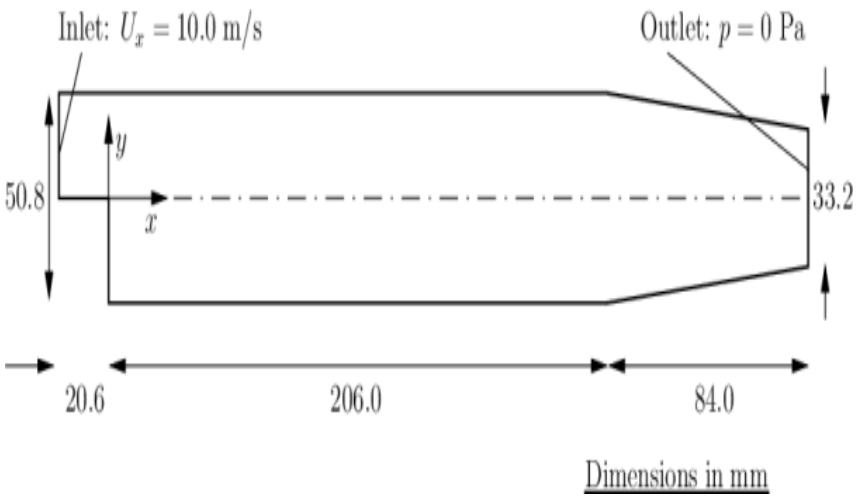


Figure 1: The backward facing step geometry.

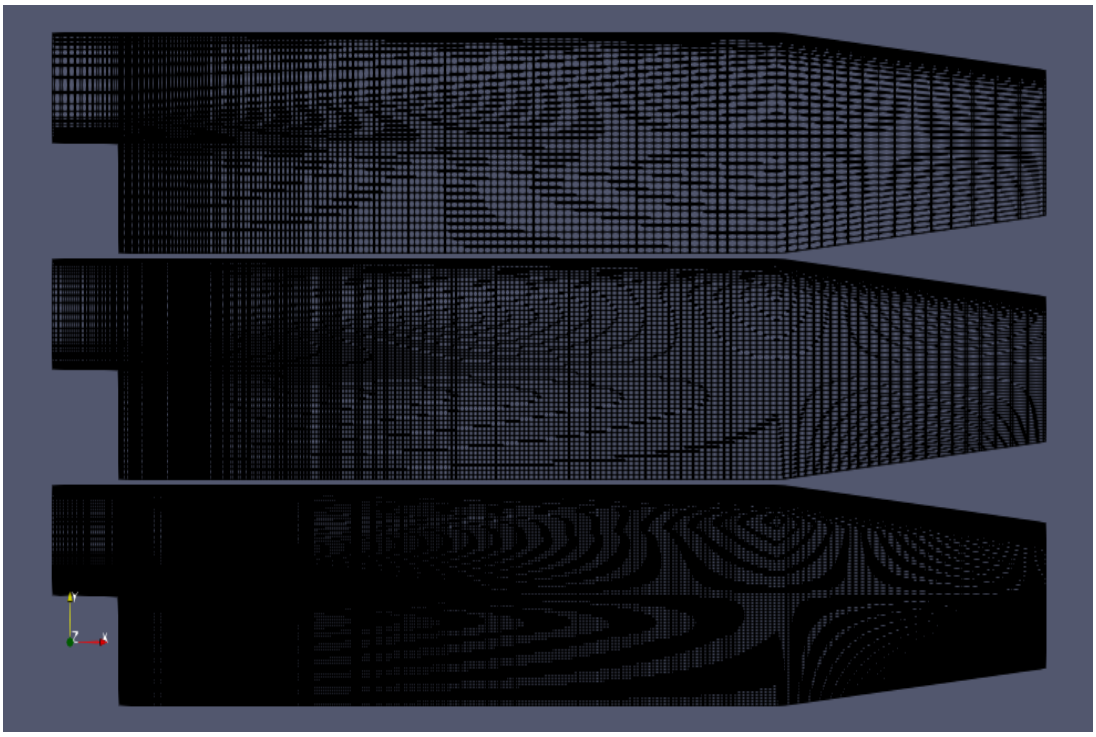


Figure 2: Different mesh sizes used in the study.

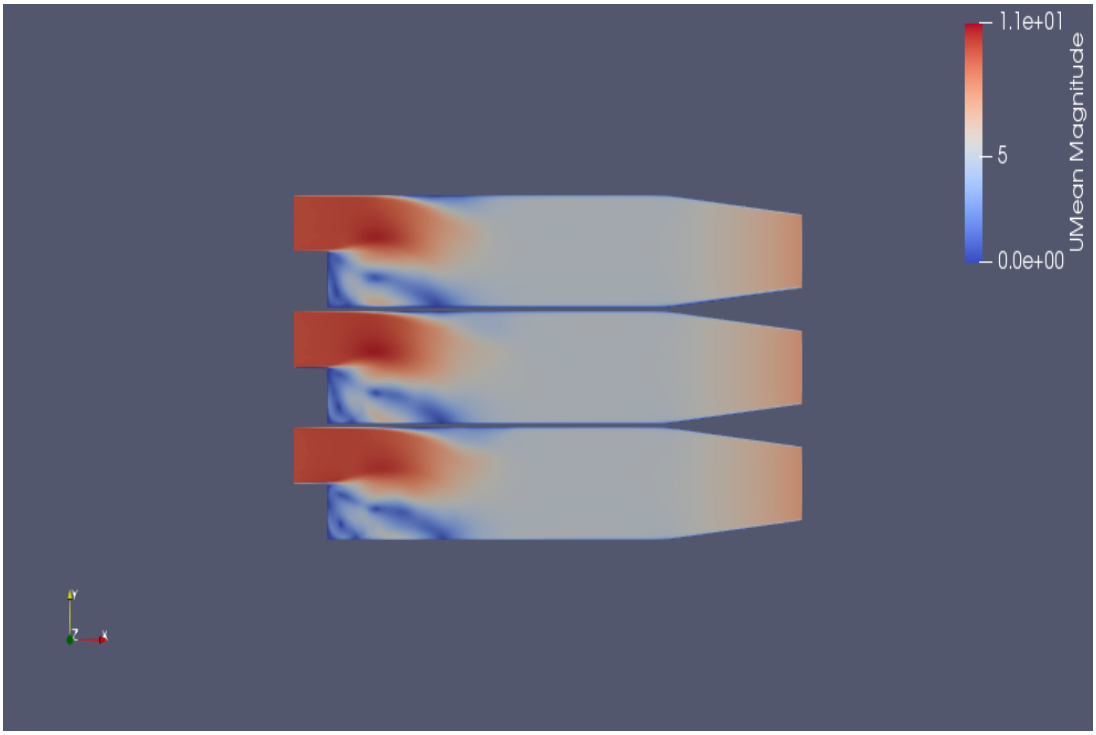


Figure 3: Mean velocity values.

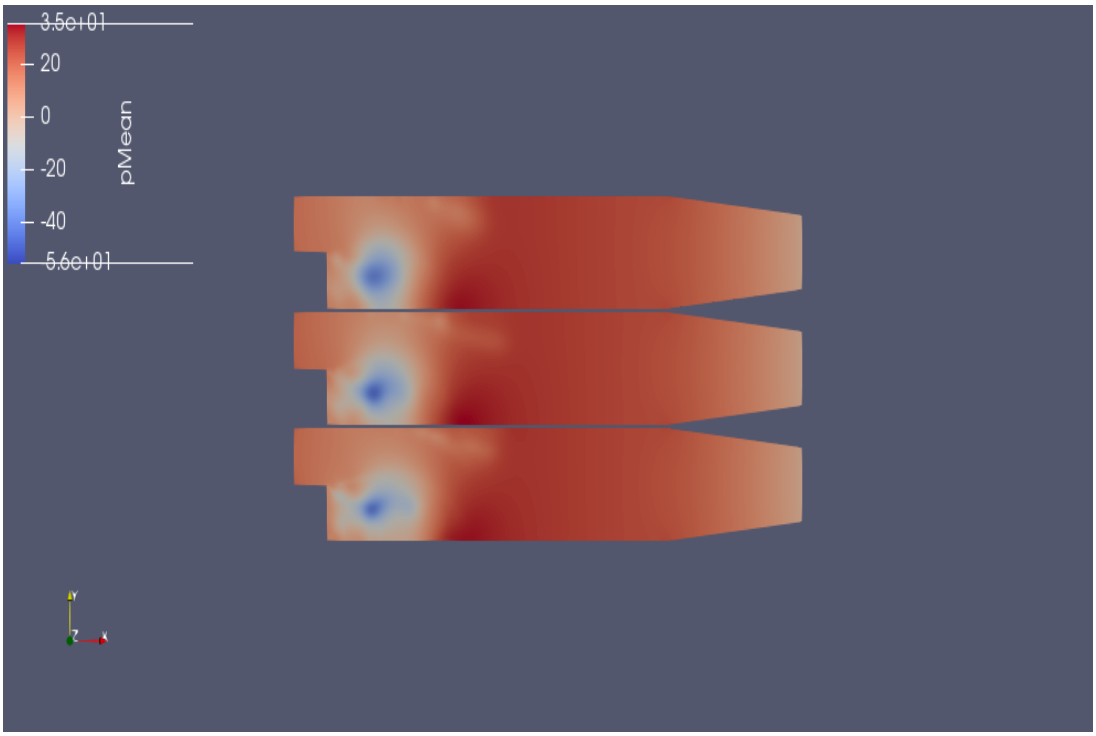


Figure 4: Mean pressure values.

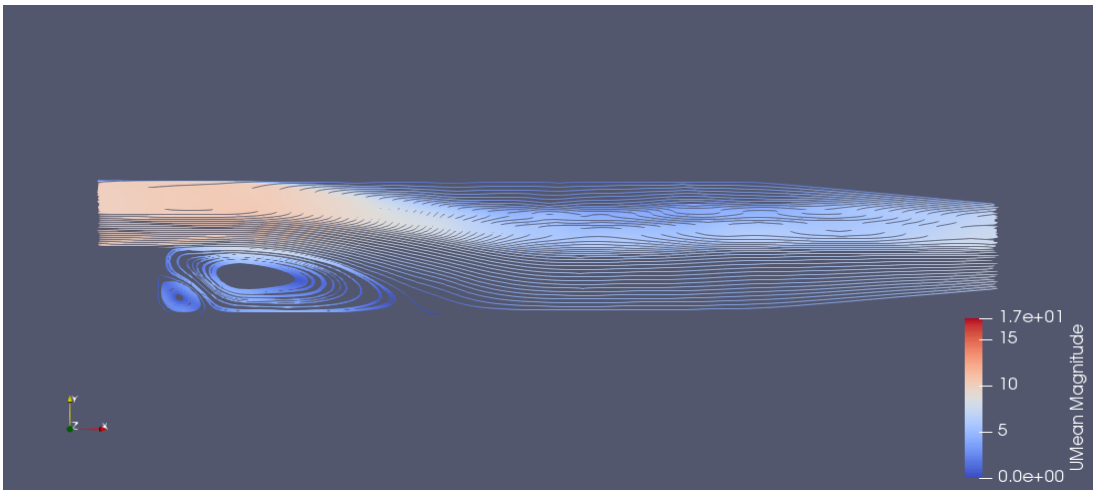
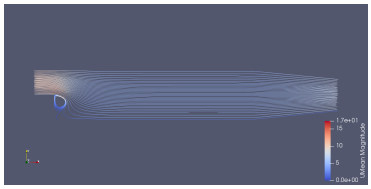
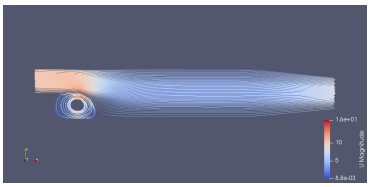


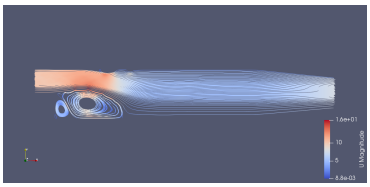
Figure 5: Streamlines for the mean velocity.



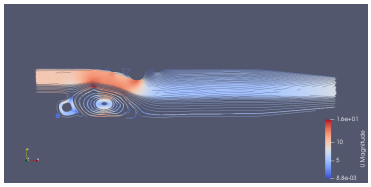
(a) Streamlines at t=0.005 s



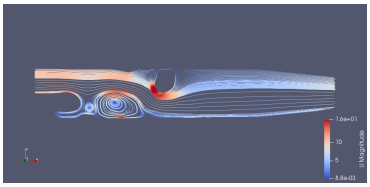
(b) Streamlines at t=0.012 s



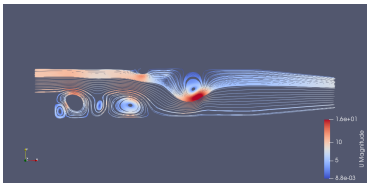
(c) Streamlines at t=0.026 s



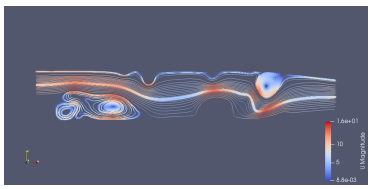
(d) Streamlines at t=0.036 s



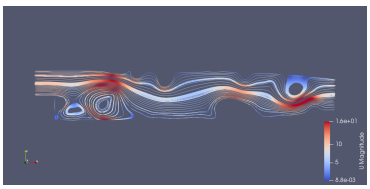
(e) Streamlines at t=0.046 s



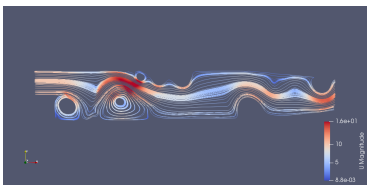
(f) Streamlines at t=0.051 s



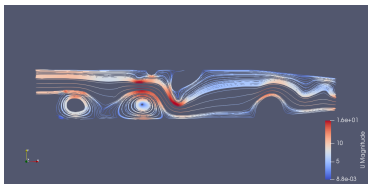
(g) Streamlines at t=0.076 s



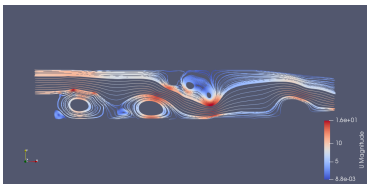
(h) Streamlines at t=0.081 s



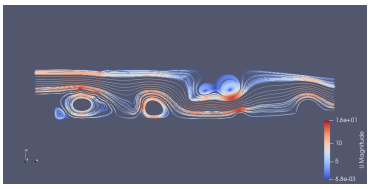
(i) Streamlines at t=0.086 s



(j) Streamlines at t=0.091 s



(k) Streamlines at t=0.096 s



(l) Streamlines at t=0.1 s

Figure 6: Streamlines development over time.